1.2.2 ▶ ▶ ▶ ▶ $F = dp/dt$

This section concerns the definition of force as the rate of change of momentum. The concept is described in three ways, and each section is marked with the appropriate color to denote its nuance.

▶ First, let us consider the most famous representation of Newtons Second Law for a closed system. Namely, that

$$\mathbf{F}_{net} = m\mathbf{a}. \tag{1.24}$$

Next, we consider the classical definition of linear momentum

$$\mathbf{p} = m\mathbf{v}. \tag{1.25}$$

Next, let us consider the rate of change of the linear momentum $\mathbf{p}$.

$$\frac{\Delta \mathbf{p}}{\Delta t} = m\frac{\Delta \mathbf{v}}{\Delta t} = m\mathbf{a} = \mathbf{F} \tag{1.26}$$

Thus, the force on an object equals the rate of change of its momentum.

▶ ▶ Now, let us use calculus to expand on the previous derivation. Again, we take the rate of change of the linear momentum of a particle.

$$\mathbf{F} = \frac{dp}{dt} = \frac{d}{dt}(m\mathbf{v}) = \frac{dm}{dt}\mathbf{v} + \frac{d\mathbf{v}}{dt}m \tag{1.27}$$

Where the product rule has been used, because we are no longer assuming that the system is closed (i.e. mass can enter and exit). We see that this adds a correction term to $\mathbf{F} = m\mathbf{a}$. Hence, we see that F must not always equal ma, which is in fact the case in some systems.

▶ Now, let us add yet another layer of nuance. We know from special relativity that the force applied on an object is equal to

$$\mathbf{F} = \gamma^3 m\mathbf{a}. \tag{1.28}$$

We get this equation by using $F = dp/dt$ on the equation for relativistic momentum. The significance of this equation is the fact that $F = ma$ is an approximation for small velocities and is not a fundamental "law" ★